

From Gigapixels to Bacteria: A Data-Centric Active Learning Framework for *Helicobacter pylori* Detection in H&E Slides

Rohan Sanjay Patil, Vidya Padmanabha, Harsha Sathish

Abstract

Helicobacter pylori is a spiral-shaped bacterium classified as a Class I carcinogen that causes chronic gastritis and gastric cancers in animals. Detecting *H. pylori* in gastric biopsies requires identifying 2–5 μm bacteria within gigapixel whole slide images (WSIs). Manual examinations are time consuming and prone to observer fatigue. While automated models offer a potential solution, standard approaches face a fundamental problem where pathologists typically annotate only a fraction of present bacteria, rendering standard object detection metrics unreliable. We present a data-centric active learning framework that addresses this incomplete annotation challenge through iterative expert-in-the-loop dataset enrichment. Our pipeline combines a hotspot identification preprocessing step to reduce the gigapixel search space, YOLOv8 for high recall discovery across active learning cycles, and Faster R-CNN for final discriminative deployment on Hematoxylin and Eosin (H&E) stained slides. We formalize Impact Precision, a clinically interpretable metric defined as the fraction of AI predictions confirmed by an expert, designed specifically for partial annotation scenarios where standard mean average precision (mAP) fails. Through three active learning (AL) iterations of training, expert review, and data refinement, the dataset grew richer with both definite positives and negatives. This improved YOLO's Impact Precision from 29.6% to 75% and transition to Faster R-CNN achieved 79.3% at clinical thresholds, reducing pathologist review burden from gigapixel scanning to verification of high-confidence candidate regions per slide. We release the complete pipeline to support reproducibility in medical imaging annotation workflows. Code: https://github.com/thws-mai/PM25_Helicobacter

1 Introduction

Helicobacter pylori species colonize the gastric mucosa of both humans and animals. While *H. pylori* diagnosis in humans is well established, *Helicobacter* infections in companion animals present unique diagnostic challenges. These bacteria are associated with chronic gastritis, but infection does not always lead to clinical symptoms. This work focuses exclusively on veterinary biopsy specimens.

Manual detection is not straightforward. Pathologists must locate minute bacteria (2–5 μm) within gigapixel WSIs, a task taking approximately 30 minutes per slide. This results in observer fatigue and potentially missed detections.

Related Work and Limitations. *H. pylori* detection in human gastric biopsies has been very successful. Klein et al. [1] demonstrated high sensitivity using deep Convolutional Neural Networks (CNNs) on immunohistochemical (IHC) stains, which are easier to process than standard H&E. While IHC staining is effective, H&E remains the diagnostic standard in veterinary pathology due to its lower cost and wider availability, and the absence of a validated IHC protocol for companion animal *Helicobacter* species.

In the veterinary domain, a prior study on this specific dataset [2] proposed a hierarchical system combining template matching and YOLO detection. While establishing feasibility, that work relies on standard precision/recall metrics which, as we demonstrate, are unreliable when ground truth annotations are sparse and partial.

The problem of incomplete annotations is well documented in medical object detection [3]. Standard detectors treat unlabeled regions as “true negative,” penalizing models for valid detections.

Our Approach. We address this challenge through an active learning pipeline developed in collaboration with an expert reviewer. Using 20 gastric biopsy WSIs, we iteratively train models, deploy them for prediction, and incorporate expert verification. Our contributions are:

- We propose an **Active Learning loop** utilizing hard negative mining [4], a technique where the model is specifically retrained on its most difficult false positives to refine its decision boundary, distinguishing bacteria from confusing artifacts such as ink marks and tissue folds.
- We formalize **Impact Precision**, a deployment-oriented metric that evaluates model utility under incomplete ground truth, where standard mean average precision (mAP, specifically mAP50) fails.
- We demonstrate that this iterative process discovers 1,456 unmarked bacteria, improving Impact Precision from 29.6% to 75% during the active learning phase and 79.3% in our final model deployment.

2 Methodology

Our system comprises three stages: (1) preprocessing to extract diagnostically relevant regions, (2) iterative active learning with expert verification to enrich the dataset, and (3) final model training with confidence-based deployment strategies. This section describes each component.

2.1 Preprocessing: Hotspot Identification

Processing a gigapixel WSI (approximately 100,000 patches) at 40 $\times$ magnification shows that most patches contain non-informative regions. We implemented a classifier-based hotspot extraction pipeline using QuPath [5], a bioimage analysis platform, which was selected for its open-source availability, established use in digital pathology research, and native support for scripted headless processing of WSIs.

Rather than simple tissue-background thresholding, we trained a Random Forest pixel classifier to segment WSIs into three classes: Glandular Epithelium, Stroma, and Background. Since *H. pylori* primarily colonizes the mucus layer and gastric gland surfaces, we designate glandular regions as high priority “Hotspots”. Fig. 1 shows this binary differentiation: mucosal interface and glandular epithelium

(red) versus inert stroma and background (green/gray). This masking forces the detector to focus on morphological distinctions of bacteria within the mucosal environment.

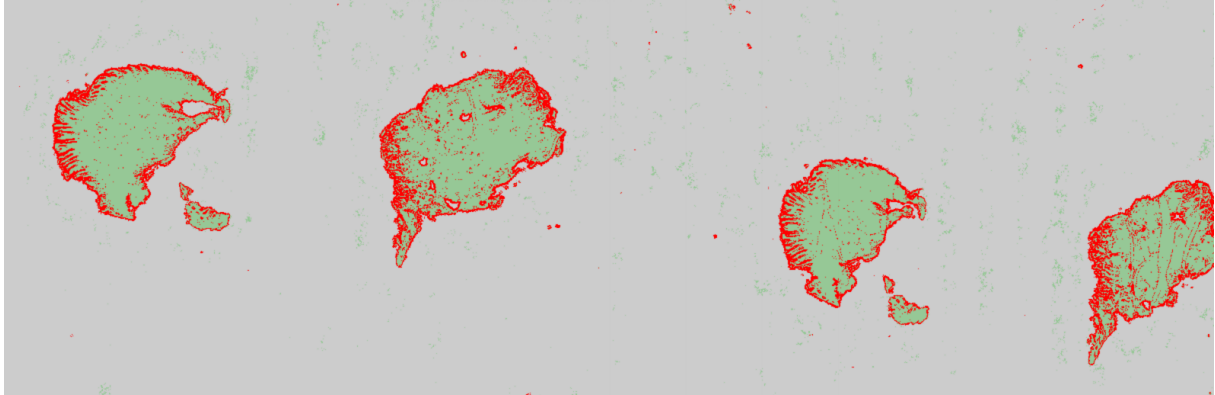


Fig. 1: The Hotspot Mask. Pathogenic glandular regions (red) occupy a small percentage of the WSI; uninformative stroma and background (green and grey) occupy the majority.

We implemented sliding window extraction as shown in Fig. 2 (512×512 pixels, 10% overlap) with a 20% minimum hotspot coverage threshold: a patch is retained only if at least 20% of its area contains glandular tissue. This heuristic reduced the search space by approximately 60% (from $\sim 40,000$ potential patches to $\sim 11,000$ per slide), significantly improving computational efficiency.

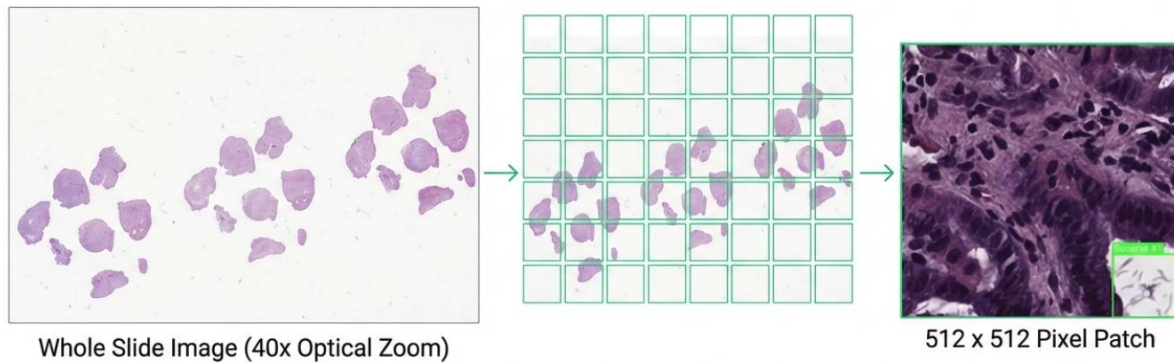


Fig. 2: WSI to patch conversion pipeline: the gigapixel slide is segmented; hotspot patches are extracted via a sliding window, and only patches with sufficient glandular coverage are retained for detection.

2.2 The Active Learning Loop

Active learning enables iterative improvement of the model by incorporating expert feedback on model predictions. These verifications are then integrated into the training set to retrain an improved model. The loop consists of the following steps:

1. **Initial Training:** Train YOLOv8 using all annotated bacteria plus randomly sampled tissue as provisional negative samples. (Initial dataset: 1,153 positive samples; no verified negatives.)

2. **Deployment:** The trained model annotates all patches unused for training (from 20 WSIs), generating predictions on approximately 2% of patches.
3. **Expert Verification:** The expert, a trained veterinary pathology collaborator with experience in gastric biopsy interpretation, reviews model predictions in collaboration with the authors and classifies them into true positives (verified bacteria) and false positives (hard negatives). True positives are added to the training set as positive samples; false positives are added as hard negatives.
4. **Dataset Enrichment:** A new enriched dataset is created by combining original annotations, newly verified true positives, and hard negatives. Provisional negatives from the initial iteration are removed to maintain dataset quality. A new YOLOv8 model is then trained.

The cycle continues until the model achieves clinically acceptable precision and recall. We denote the baseline pre-active-learning state as AL 0, and each subsequent active learning iteration as AL 1, AL 2, and AL 3.

2.3 The Models

We employed a two-model strategy tailored to the distinct requirements of active learning and clinical deployment.

YOLOv8 for Active Learning Cycles. For active learning, we needed a model that could find as many potential bacteria as possible. YOLOv8 [6] was selected because its single-stage architecture generates high recall detections, enabling detection of the maximum number of possible bacteria across the full 11,000-patch-per-slide search space. We initialized YOLOv8s with COCO [7] pre-trained weights and trained 150 epochs with aggressive augmentations to handle staining variations and orientation diversity (detailed specifications in Appendix B).

Faster R-CNN for Final Deployment. For deployment, we used Faster R-CNN, which provides more discriminative detections through its two-stage refinement process. The region proposal network first identifies candidate regions, then a classification head evaluates each proposal carefully to distinguish bacteria from morphologically similar artifacts. We used a ResNet-50-FPN backbone with anchor sizes tuned to bacterial morphology at 40 $\times$ magnification (see Appendix B).

3 Evaluation of Partial Annotations

3.1 The Incompleteness Problem

In WSI scale histopathology, traditional object detection metrics assume complete ground truth: all instances of the target class are annotated. This assumption fails for diagnostic annotations. A pathologist confirming *H. pylori* infection may label 50 bacteria to make a diagnosis, then proceed to the next slide, leaving 500+ bacteria unlabeled. This creates two types of unmatched predictions:

- **True Errors:** Ink artifacts, tissue debris, dust
- **Unlabeled Positives:** Genuine bacteria missed by the pathologist.

Standard metrics (mAP, precision) cannot distinguish these cases, systematically penalizing the model for detecting bacteria the pathologist did not annotate.

3.2 Impact Precision: A Clinical Metric

We formalize Impact Precision (P_{impact}) as a practical reinterpretation of standard precision, specifically adapted for expert-in-the-loop deployment in partial annotation scenarios. While mathematically equivalent to precision, P_{impact} specifically accounts for the clinical utility of an AI system where the ground truth is known to be incomplete, shifting the focus from 'matching labels' to 'confirming pathology'. It is given by:

$$P_{\text{impact}} = \text{Expert - confirmed bacteria} / \text{Total AI predictions} \quad (1)$$

For example, if $P_{\text{impact}} = 75\%$ that means that when the AI system flags a region, there is a 75% probability that a pathologist will confirm the presence of *H. pylori*. This directly determines whether AI assistance reduces or increases pathologist workload.

P_{impact} does not directly quantify recall (sensitivity), as we cannot determine what fraction, of all the bacteria from the WSI were detected. In medical diagnostics, missing a true positive (false negative) carries a higher clinical cost than a false alarm [8]. It is therefore necessary to clarify the role of recall in our system. Our system is explicitly designed as a screening assist tool which reduces the area a pathologist must manually examine, from a full gigapixel WSI to a ranked list of high confidence patches. The pathologist examines the flagged regions. The AI does not replace this step but helps reduce the region of examination. Recall, while not fully measurable due to incomplete ground truth, is addressed structurally since the hotspot preprocessing retains all glandular regions, and the active learning loop systematically discovers previously missed bacteria across multiple iterations. High impact precision at deployment is therefore not a decision to sacrifice recall but a decision to reduce the false alarm rate.

By prioritizing P_{impact} over mAP throughout active learning, we discovered 1,456 unmarked bacteria across 20 WSIs while systematically improving from 29.6% to 75% Impact precision.

3.3 The Metric Paradox

Our results expose a flaw in traditional metrics for partial annotation scenarios. Table 1 presents mAP, YOLO precision (computed against partial annotations), and Impact Precision (computed via expert verification at corresponding YOLO detection confidence thresholds) across iterations.

Table 1: Metric Divergence Under Incomplete Annotations

Iteration	mAP	YOLO Precision	$P_{\text{impact}} @0.15$	$P_{\text{impact}} @0.25$
AL 0	0.41	0.70	29.6%	38%
AL 1	0.48	0.73	59.4%	90%
AL 2	0.35	0.42	14.5%	15%
AL 3	0.33	0.40	66.8%	75%

Traditional metrics suggested marginal improvement: mAP increased by only 7 points (0.41 $\rightarrow$ 0.48) from AL 0 to AL 1, suggesting modest gains. However, expert verification revealed that clinical utility more than doubled. The model learned to discriminate bacteria from artifacts, but this improvement is invisible to mAP because newly discovered bacteria are treated as false positives. This divergence is visualized in Fig. 3.

More strikingly, naively selecting the model with highest mAP (AL 1, mAP 0.48) would yield merely good performance, while selecting the model with lowest mAP (AL 3, mAP 0.33) yields

excellent performance (66.8% Impact Precision). mAP and standard precision are thus unreliable indicators of clinical utility under incomplete annotations.

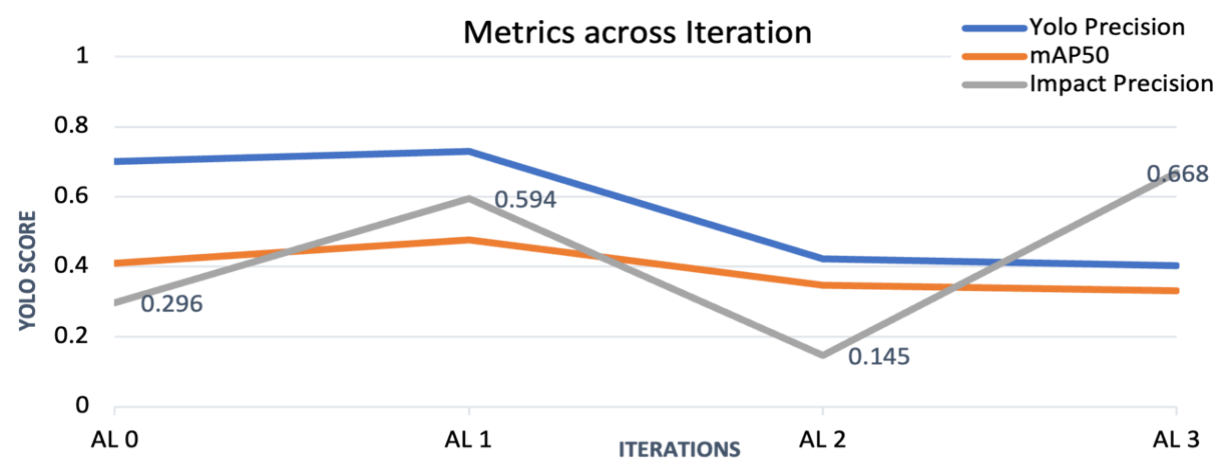


Fig. 3: Traditional metrics suggest inconsistent, non-monotonic progress, while Impact Precision correctly identifies AL 3 as the superior model despite its lower mAP score.

4 Dataset Evolution Through Active Learning

Three active learning cycles evolved our detection system, each revealing insights that shaped our data engineering strategy. This section presents the evolution of a reproducible workflow for researchers facing similar partial annotation challenges. Table 2 summarizes the progression accumulating each layer over the previous one.

Table 2: Dataset Growth and Performance Across Active Learning Iterations

Iteration	Positives	Hard Neg*	Total	P_impact**	Key Insight
AL 0	1,153	0	1,153	29.6%	Ink artifacts
AL 1	2,150	2,372	4,522	59.4%	Conservative
AL 2	2,412	2,551	4,963	14.5%	Ratio failure
AL 3	2,609	3,708	6,317	66.8%	Optimal

*Accumulated verified negatives at end of each iteration. AL trainings used fewer randomly sampled negatives (ex: 4:1 ratio experiment), not the full accumulated pool.

Note: The Hard Neg column for AL 2 (2,551) represents the total accumulated pool at iteration end; only 603 of these were used in AL 2 training (4:1 ratio experiment).

**P_impact at confidence threshold 0.15. See Table 1 for @0.25 values.

4.1 Iteration 0: Baseline and Artifact Discovery

Initial deployment. We began with 1,153 expert annotated positives from 20 WSIs (Table 2). Critically, no verified negative examples existed as the dataset contained only positive samples and randomly sampled tissue regions serving as provisional negatives. This asymmetry would prove consequential.

We trained a baseline YOLOv8 model and deployed it on unannotated regions of the 20 WSI to discover missed bacteria and this acts as data mining rather than evaluation and is the intended purpose of active learning. A small subset of initial ground-truth annotations from the 2 held out slides (~50–100 instances, <2% of the final 6,317-sample training set) was present in the baseline AL 0 training data, as the full hold-out protocol was established after initial annotation. Processing approximately 120,000 patches generated 3,369 candidate detections.

Expert verification revealed that 2,372 of 3,369 detections were ink artifacts from slide preparation, achieving only 29.6% Impact Precision. Fig. 4 illustrates the morphological similarity between ink artifacts and genuine bacteria that caused these systematic false positives.

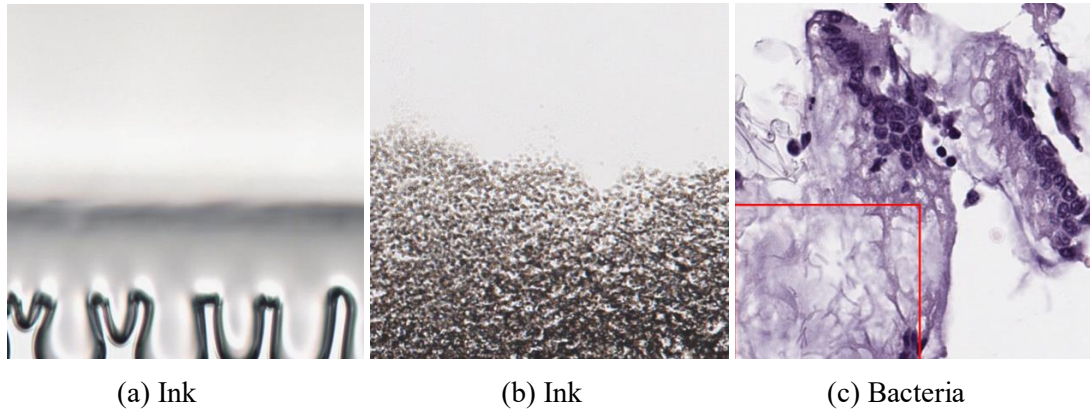


Fig. 4: Morphological Discrimination Challenges. (a, b) Ink artifacts falsely detected as bacteria due to visual shape similarity. (c) A faded bacterium correctly identified in Iteration 1.

Which confirmed that without negative examples, the model’s decision boundary was defined solely by positive instances and any dark, curved structure resembling spiral bacteria triggered detection. The model learned “what bacteria look like,” but not “what bacteria are NOT.”

Rather than discarding these 2,372 failures, we recognized their diagnostic value. These were not random errors, they were systematically difficult cases where visual similarity to bacteria was highest, providing maximal information gain for learning discriminative boundaries [4]. This decision to include the failures doubled the dataset (1,153 → 4,522 samples) and gave the model a discriminative boundary to learn.

4.2 Iteration 1: Conservative Behavior

We retrained YOLOv8 with approximately 1:1 positive-to-negative ratio: 2,150 positives (1,153 original + 997 newly discovered) and 2,372 hard negatives. The hypothesis was that the model would learn to discriminate bacteria from ink artifacts while maintaining detection sensitivity.

Deployment produced 441 candidates (87% reduction), with 262 verified bacteria (59.4% Impact Precision). Ink artifacts dropped dramatically, but the model became overly conservative, missing genuine bacteria; in doing so it had prioritized precision over recall. That meant that confidence scores were meaningful. All detections above 0.5 confidence were correct, and 90% of detections above 0.25 confidence were verified bacteria. The model had become reliable, even if it under detected overall.

4.3 Iteration 2: The Recall Recovery Experiment

To recover recall, we adjusted the training ratio to 4:1 (2,412 positives, 603 randomly sampled negatives from the definite negative pool). If a 1:1 ratio made the model too cautious, a positive heavy ratio should shift it toward more detections.

Detections rebounded to 1,354, but only 197 were genuine bacteria (14.5% Impact Precision) indicating that the model had memorized a subset of ink rejection without learning general discrimination. It reverted to aggressive detection, generating 1,157 new false positives. Importantly, these were different artifacts than in Iteration 0: the model retained its ink rejection learning but failed to generalize to other artifact types (tissue folds, cellular debris, staining irregularities).

The positive-to-negative ratio is a critical hyperparameter in hard-negative mining. A 4:1 ratio allows the model to memorize specific artifact rejection without learning general discriminative principles; the 1:1 ratio in Iteration 1 was closer to optimal, though it sacrificed recall.

Despite poor precision performance, Iteration 2 provided valuable data. Each active learning cycle expands not only the count of verified bacteria, but the diversity of hard negatives, progressively teaching the model more subtle boundaries. The cumulative dataset growth is visualized in Fig. 5. By iteration 2's conclusion, the dataset grew from 1,153 samples to 6,317 samples (2,609 positives, 3,708 negatives) more than a 4-fold growth after AL2 deployment.

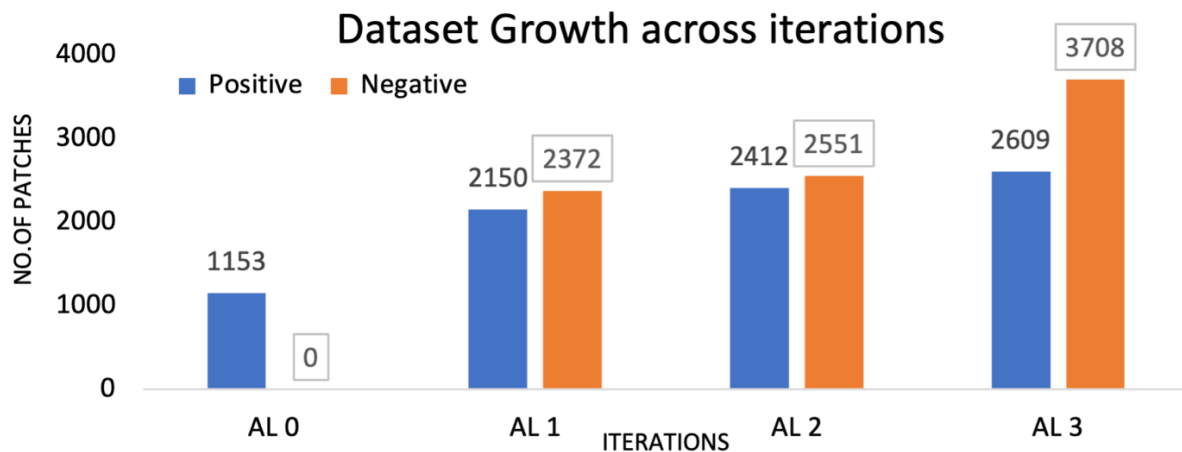


Fig. 5: Dataset evolution across active learning iterations. Each cycle expanded both positive (blue) and hard negative (orange) examples, creating a progressively more comprehensive training set.

4.4 Iteration 3: Restoring the Optimal Ratio

Having identified that the 4:1 positive heavy ratio caused generalization failure in Iteration 2, we decided to consider the complete verified patches (2,609 pos and 3,708 neg) forming about 2:3 ratio. Crucially, the negative pool was now far richer than in any iteration with 3,708 verified hard negatives covering ink artifacts, tissue folds, cellular debris, and staining irregularities, compared to only 2,372 ink focused negatives used in iteration 1.

Deployment produced 437 candidates, with 292 verified genuine bacteria (66.8% Impact Precision at threshold 0.15; 75% at threshold 0.25). This represents the best balance achieved across all YOLO iterations where impact precision nearly recovered to iteration 1 levels, while the negative pool was now

diverse enough to enable generalization across artifact types rather than memorizing ink specific patterns. Meaning a comprehensive, verified dataset with diverse hard negatives outperforms any ratio-optimized training set built from narrow negatives. Each active learning cycle earns its value not primarily through more data, but through more diverse hard negatives that progressively sharpen the decision boundary.

5 Final Deployment Analysis: Transition to Faster R-CNN

A planned dual-model strategy. The two-model design was not a response to YOLO underperforming; it was an architectural decision. To determine clinically viable confidence thresholds for deployment, we analyzed how threshold values affect the precision-recall trade off across the range 0.10–0.95, shown in Fig. 6. This analysis revealed distinct reliability zones that inform our deployment strategy.

The active learning phase must optimize for recall above all else; missing bacteria is irreversible damage to the dataset. YOLOv8’s single-stage, recall-biased architecture is the correct tool for this phase by design. For clinical deployment, the error cost inverts completely. Every flagged region represents direct pathologist effort, and false positives now carry a concrete workload cost. Precision becomes the primary objective. A single model cannot serve both phases without compromising one: high recall during AL conflicts with high precision at deployment. This is precisely why Faster R-CNN was reserved for the final model.

Its two-stage architecture, a region proposal network followed by a dedicated classification head is built for discriminative precision: each candidate region receives an independent, careful evaluation before a prediction is committed, rather than a single-pass grid decision. Deploying Faster R-CNN during the active learning cycles would also have been technically counterproductive. The region proposal network requires diverse negative examples to generate meaningful candidate regions; the active learning Iteration 0 dataset contained zero verified negatives. Training Faster R-CNN at that stage would have produced unreliable proposals and systematically excluded genuine bacteria from the enrichment loop; the opposite of what active learning requires.

By the end of active learning 3, the dataset had grown to 6,317 samples with 3,708 diverse hard negatives covering ink artifacts, tissue folds, cellular debris, and staining irregularities. This breadth is precisely what Faster R-CNN’s classification head needs to learn reliable discrimination.

While YOLOv8 excelled at rapid iteration and dataset enrichment, the model struggled with hard negatives specifically tissue artifacts which are visually similar to bacteria but biologically distinct. After training Faster R-CNN on the final enriched dataset, we evaluated its performance by testing how confidence thresholds affect detection quality. Evaluation was performed on two WSIs held out from all active learning iterations (see Section 4.1 for full evaluation protocol).

At very low thresholds (0.10–0.30), the model approaches comprehensive detection, finding 377 of 398 verified bacteria (94.7% recall on the verified set) but generating 1,172 total predictions. Only 377 are confirmed bacteria; the remaining 795 are false alarms. This floods pathologists with artifacts, defeating the purpose of automation.

As the threshold increases, a clinically practical operating range emerges around 0.60–0.80. At threshold 0.75, predictions drop to 362 while still catching 287 bacteria, filtering out 885 false alarms.

At threshold 0.60, the system flags 559 regions, but only 67% are actual bacteria meaning 1 in 3 flagged regions is a false alarm. At threshold 0.75, only 362 regions are flagged and 79.3% are real bacteria meaning only 1 in 5 is false. This substantially reduces pathologist workload while improving precision.

Table 3 summarizes the final performance of both models. The final YOLOv8 model (at threshold 0.25) achieves 79% recall on the verified set but with a critical limitation: its conservative behavior after hard negative training means it flags fewer total regions, missing bacteria that Faster R-CNN detects. Faster R-CNN at threshold 0.75 achieves 79.3% Impact Precision indicating approximately 4 in 5 flagged regions contain confirmed bacteria. At the cost of 7% recall compared to YOLOv8 (72% vs 79%), reducing total detections from 437 to 362 meaning lesser pathologist workload. The two models serve complementary roles: YOLOv8 for high-recall discovery during active learning, Faster R-CNN for high precision clinical deployment. This shifts the pathological workflow from exhaustive manual scanning to efficient verification of AI flagged regions. For final clinical deployment/evaluation Faster R-CNN was assessed on two hold-out WSIs from a separate 22-slide cohort withheld almost entirely from active learning.

Table 3: Model Comparison at Deployment Thresholds

Model	Threshold	P_impact	Recall	Detections
YOLOv8 (final)	0.25	75%	79%	437
Faster R-CNN	0.60	67%	~95%	559
Faster R-CNN	0.75	79.3%	72%	362
Faster R-CNN	0.85	>90%	~50%	<200

Recall computed against the verified positive set of 398 bacteria.

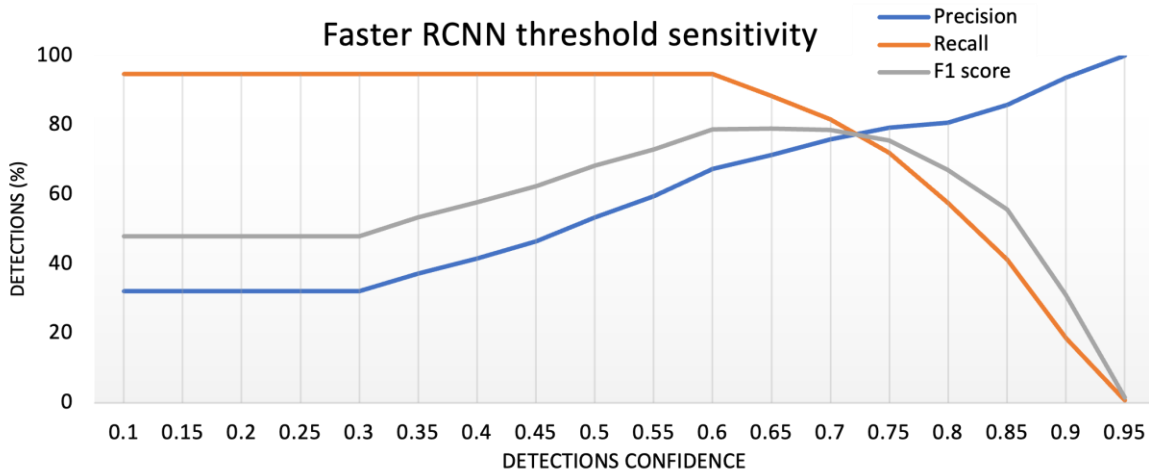


Fig. 6: Faster R-CNN Impact Precision, recall, and F1 score versus confidence threshold. The region (0.60–0.80) indicates the clinically practical operating range.

6 Limitations and Future Work

- **Annotation incompleteness and recall quantification.** A fundamental constraint of this work is that true recall; the fraction of all bacteria in a WSI that the system detects cannot be quantified with the current dataset. Despite active learning improvements, our annotations remain incomplete. Some false negatives may be genuine missed detections that future iterations would reveal. A prospective clinical trial with fully annotated reference slides remains the necessary next step to quantify sensitivity directly and validate Impact Precision against a complete ground truth. Researchers building on this work who have access to exhaustively annotated datasets even a single slide would contribute meaningfully.
- **Single stain focus.** The system was developed and evaluated exclusively on H&E-stained slides from a single veterinary diagnostic laboratory using a consistent tissue preparation protocol. Generalizability to slides prepared at different institutions, digitized on different scanner hardware, or processed with different reagent batches remains unvalidated. Even within H&E staining, inter-laboratory variation in staining intensity and tissue fixation can meaningfully alter the appearance of bacterial morphology and surrounding tissue, which the current augmentation strategy only partially addresses. Extending the pipeline to alternative staining protocols such as Giemsa or Warthin-Starry, both used in veterinary *Helicobacter* diagnosis would increase clinical utility but would require retraining with protocol-specific annotations.
- **Patch overlap and Integration with laboratory systems.** The sliding window extraction uses a 10% overlap between adjacent patches, which introduces a potential source of data leakage. Since the train/evaluation split was performed at the patch level rather than the slide level, spatially adjacent patches sharing pixel content could appear in both training and evaluation sets. While the scale of this overlap is modest and unlikely to materially alter deployment conclusions given the large disparity between training and evaluation set sizes, it represents a methodological impurity that future work should address. Full clinical deployment requires interfacing with laboratory information systems for automated report generation and result tracking.

7 Conclusion

This work demonstrates that in medical object detection under incomplete annotations, data quality is more influential than model complexity. Clean, iteratively verified labels improved detection performance far beyond what architectural changes alone could achieve. Three broader conclusions emerge.

First, active learning with hard negative mining is an effective strategy for the common medical imaging scenario where initial annotations are sparse and partial: each iteration enriches both the quantity and diversity of training examples. Second, standard metrics such as mAP are unreliable indicators of clinical utility when ground truth is incomplete; Impact Precision, formulated as the expert-confirmed fraction of AI predictions, provides a deployment-oriented alternative that directly measures whether the system reduces pathologist workload. Third, the complete pipeline from hotspot preprocessing, iterative dataset enrichment, to confidence-threshold deployment is fully reproducible and applicable to analogous detection problems in histopathology where exhaustive annotation is impractical.

For pathologists, the system transforms gigapixel slide examination from exhaustive manual scanning into efficient verification of a ranked list of high-confidence candidate regions. The active learning loop is not considered finished: further iterations with additional slides would continue improving both recall and precision, as neither the model nor the ground truth is ever truly complete.

Acknowledgement: This work was supervised by Prof. Dr. Magda Gregorová, from the Technical University of Applied Sciences Würzburg-Schweinfurt (THWS). We sincerely thank LABOKLIN for providing the veterinary gastric biopsy dataset and expert pathological annotations. Special thanks to Philipp Ockermann for his essential support throughout the active learning process as the expert. THWS is acknowledged for providing computational resources.

References

- [1] S. Klein, A. Gildenblat, A. Khader, et al., Deep learning for sensitive detection of *Helicobacter pylori* in gastric biopsies. *BMC Gastroenterology*, 2020 Dec 11;20(1):417. doi: 10.1186/s12876-020-01494-7. PMID: 33308189; PMCID: PMC7731757.
- [2] N. Sharma, Z. Lu et al., A Computer-Aided Detection System for *Helicobacter pylori* in Gastric Biopsy Slides. [Unpublished Manuscript] THWS, Previous Project Report / using LABOKLIN Dataset, 2025.
- [3] H. Bilen and A. Vedaldi, Weakly supervised deep detection networks. *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 2846–2854, 2016.
- [4] A. Shrivastava, A. Gupta, and R. Girshick, Training region-based object detectors with online hard example mining. *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 761–769, 2016.
- [5] P. Bankhead, M. B. Loughrey, J. A. Fernández, et al., QuPath: Open-source software for digital pathology image analysis. *Sci Rep.* 2017 Dec 4;7(1):16878. doi: 10.1038/s41598-017-17204-5. PMID: 29203879; PMCID: PMC5715110.
- [6] G. Jocher, A. Chaurasia, and J. Qiu, Ultralytics YOLOv8. <https://github.com/ultralytics/ultralytics>, version 8.0.0, 2023.
- [7] T.-Y. Lin, M. Maire, S. Belongie, et al., Microsoft COCO: Common Objects in Context. *European Conference on Computer Vision (ECCV)*, pp. 740–755, 2014. doi: 10.1007/978-3-319-10602-1_48.
- [8] S. Hicks, I. Strümke, V. Thambawita, M. Hammou, M. Riegler, P. Halvorsen and S. Parasa. On evaluation metrics for medical applications of artificial intelligence. *Sci Rep.* 2022 Apr 8;12(1):5979. doi: 10.1038/s41598-022-09954-8. PMID: 35395867; PMCID: PMC8993826.

Appendix A: Additional Data and Visualizations

A.1 Detailed Iteration Statistics

Table 4 presents comprehensive deployment statistics across all active learning iterations and deployed Faster R-CNN model. The confidence score distributions reveal how model calibration improved: Iteration 1 achieved 100% precision at confidence >0.50 , while Iteration 2’s ratio imbalance caused precision collapse and no detections at confidence >0.75 across active learning. By contrast, Faster R-CNN generated a substantially higher proportion of detections above the 0.75 confidence threshold, reflecting its superior discriminative calibration for this task. Artifact breakdowns show the shift from predominantly ink artifacts (Iteration 0) to diverse tissue debris (Iteration 2).

Table 4: Comprehensive Iteration Statistics: detection counts, Impact Precision, confidence calibration, and artifact category breakdown

Metric	Iter 0	Iter 1	Iter 2	Iter 3	RCNN
Total Detections	3,369	441	1,354	437	1172
Verified TP	997	262	197	292	377
Verified FP	2,372	179	1,157	145	795
P_impact (%) *	29.6	59.4	14.5	66.8	32.17
Conf. > 0.25 (%) **	38	90	15	75	32.17
Conf. > 0.50 (%)	42	100	25	100	53.47
Conf. > 0.75 (%) **	-	-	-	-	79.28
Ink Artifacts	High	Low	Low	Low	Low
Tissue Debris	High	Low	High	Low	Low

*The overall Precision based on complete set of detections and verifications without specific thresholding implementations.

**The preferred confidence thresholding that produces the best balance of both precision and recall for the active learning YOLO (0.25) and deployed Faster R-CNN (0.75) models.

A.2 YOLOv8 Threshold Analysis

Based on threshold analysis, the final YOLOv8 model is deployed at a confidence threshold of 0.25, achieving 75.25% Impact Precision while maintaining comprehensive recall. Fig. 7 demonstrates threshold sensitivity across iterations. The left panel shows the precision-recall trade-off for the final model: higher thresholds improve precision but sacrifice recall. The right panel compares Impact Precision across all iterations, revealing that Iterations 1 and 3 achieved high precision with conservative detection, while Iterations 0 and 2 exhibited aggressive detection with lower precision illustrating the direct impact of positive-to-negative training ratios on model behavior.

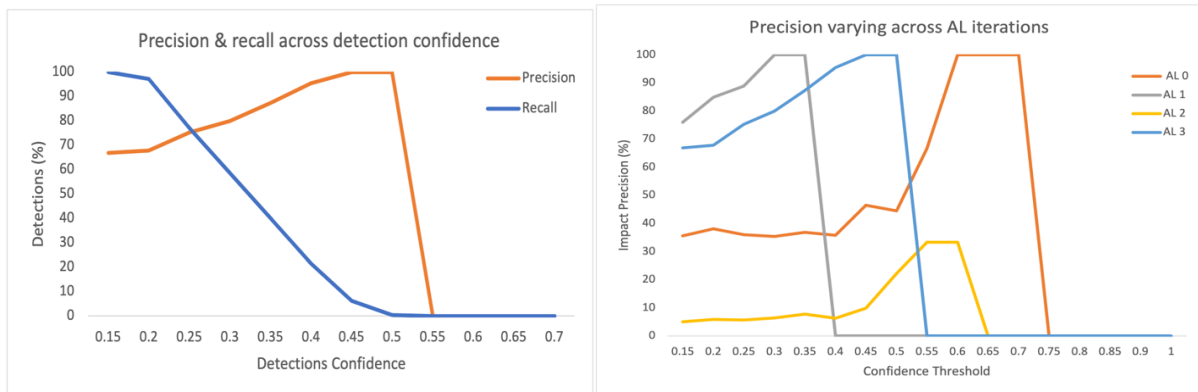


Fig. 7: YOLOv8 threshold analysis. Left: Impact Precision and recall versus confidence threshold for the final model. Right: Impact Precision comparison across all active learning iterations.

Appendix B: Model Training Specifications

YOLOv8s Configuration:

- Pre-training: COCO dataset weights
- Training: 150 epochs, batch size 16
- Optimizer: AdamW (LR: 0.001 $\rightarrow$ 0.0001)
- Hardware: NVIDIA RTX 3090

- Augmentations: Rotation ($\pm 180^\circ$), flips, HSV jitter ($H \pm 0.015$, $S \pm 0.7$, $V \pm 0.4$), scaling ($0.5\text{--}1.5\times$)
- Inference: Confidence 0.15 (active learning), 0.25 (deployment); NMS IoU 0.45

Faster R-CNN Configuration:

- Architecture: ResNet-50 + Feature Pyramid Network (FPN)
- Pre-training: ImageNet weights
- Anchor sizes: 4, 8, 16, 32, 64 pixels (tuned for 2–5 μm bacteria at $40\times$ magnification)
- Augmentations: Same as YOLOv8 for consistency
- Inference: Confidence 0.75 (clinical deployment)

ROHAN SANJAY PATIL

THWS, MAI
Applied Computer Science
Franz-Horn-Straße 2, 97082
Würzburg, GERMANY
E-mail: rohansanjay.patil@study.thws.de

VIDYA PADMANABHA

THWS, MAI
Applied Computer Science
Franz-Horn-Straße 2, 97082
Würzburg, GERMANY
E-mail: vidya.padmanabha@study.thws.de

HARSHA SATHISH

THWS, MAI
Applied Computer Science
Franz-Horn-Straße 2, 97082
Würzburg, GERMANY
E-mail: harsha.sathish@study.thws.de